

SLVS Voltage center Adjustment

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

2. Description

The configuration of SLVS voltage center adjustment is described in this document.

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3. Features

◆ SLVS voltage center is adjusted.

◆ Support mode

Normal mode (XVS / XHS sync)

Sequential Trigger mode

Fast Trigger mode

◆ Support readout drive mode

All-pixel scan mode

Vertical / Horizontal 1/2 Subsampling mode

2 × 2 FD binning mode (The product with a color filter is excluded)

ROI mode

Overlap ROI mode

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series / IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

Contents

1.	Product type	1
2.	Description	1
3.	Features	2
Contents		3
4.	Register Map	4
5.	SLVS voltage center adjustment	6
5.1.	Feature overview	6
5.2.	Register setting	6
5.3.	Sensor setting flow and sequence	7
5.3.1.	Read the register value every time standby is canceled, and adjust the SLVS voltage center.	8
5.3.2.	Adjust the SLVS voltage center with the register value read in advance.	11
6.	Note	14
Revision History		15

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4. Register Map

The register map for SLVS voltage center adjustment is shown below. Please refer to the product specification for register setup other than those list.

Chip ID = 0Bh (Write: Chip ID = 0Bh, Read: Chip ID = 8Bh, I2C: 39**h)

Address		bit	Register Name	Description	Default value after reset				Reflection timing			
4-wire	I ² C				By register		By address					
52h	3952h	0		SLVS voltage center adjustment On: Set the value by subtracting "6" from the read value of Chip ID = 8Bh, Address = E4h, bit = [4:0].	15h *1	08h *2	15h *1	08h *2	I			
		1										
		2										
		3										
		4										
		5									Fixed to 0	0
		6									Fixed to 0	0
		7									Fixed to 0	0
53h	3953h	0		SLVS voltage center adjustment On: Set the value by subtracting "6" from the read value of Chip ID = 8Bh, Address = E5h, bit = [4:0].	15h *1	08h *2	15h *1	08h *2	I			
		1										
		2										
		3										
		4										
		5									Fixed to 0	0
		6									Fixed to 0	0
		7									Fixed to 0	0
56h	3956h	0		1Eh: SLVS voltage center adjustment On 18h: SLVS voltage center adjustment Off	18h		18h	I				
		1										
		2										
		3										
		4										
		5										
		6										
		7										
58h	3958h	0		1: SLVS voltage center adjustment On 0: SLVS voltage center adjustment Off	0	00h		I				
		1		1: SLVS voltage center adjustment On 0: SLVS voltage center adjustment Off	0			I				
		2		Fixed to 0	0			-				
		3		Fixed to 0	0			-				
		4		Fixed to 0	0			-				
		5		Fixed to 0	0			-				
		6		Fixed to 0	0			-				
		7		Fixed to 0	0			-				
5Ah	395Ah	0		1: SLVS voltage center adjustment On 0: SLVS voltage center adjustment Off	0	00h		I				
		1		Fixed to 0	0			-				
		2		Fixed to 0	0			-				
		3		Fixed to 0	0			-				
		4		Fixed to 0	0			-				
		5		Fixed to 0	0			-				
		6		Fixed to 0	0			-				
		7		Fixed to 0	0			-				

^{*1} IMX530 series and IMX540 series

^{*2} IMX535 series, IMX545 series and IMX548 series

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
E4h	39E4h	0		Set the value by subtracting "6" from the read value of this register to Chip ID = 0Bh, Address = 52h, bit = [4:0]. * The value once read does not change. The value is different for each sensor. * This register is read-only and cannot be written.	Undefined	Undefined	I
		1					
		2					
		3					
		4					
		5		Fixed to 0	0		-
		6		Fixed to 0	0		-
		7		Fixed to 0	0		-
E5h	39E5h	0		Set the value by subtracting "6" from the read value of this register to Chip ID = 0Bh, Address = 53h, bit = [4:0]. * The value once read does not change. The value is different for each sensor. * This register is read-only and cannot be written.	Undefined	Undefined	I
		1					
		2					
		3					
		4					
		5		Fixed to 0	0		-
		6		Fixed to 0	0		-
		7		Fixed to 0	0		-

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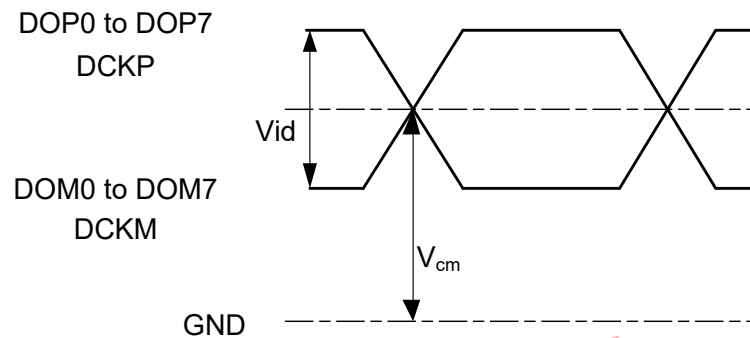
5. SLVS voltage center adjustment

5.1. Feature overview

By applying this function, the SLVS voltage center can be raised by 50 mV.

Also, the maximum value of the differential voltage is raised by 100 mV, and the current of the V_{DDL} pin increases by about 10 mA maximum.

Single end output



Definition of the characteristics of SLVS (Single end output)

Symbol	Item	Min.	Typ.	Max.	Unit	Remarks
R _o	Sensor output impedance	30	—	60	Ω	
V _{cm}	Voltage center	200	—	300	mV	*1
V _{id}	Differential voltage	140	—	400	mV	*1

*1 R_{in} = 100Ω.

5.2. Register Setting

The register settings for SLVS voltage center adjustment are shown below.

Address	bit	Register Name	Initial Value	Setting Value
Chip ID = 0Bh				
52h	[4:0]		15h	*1
53h	[4:0]		15h	*2
56h	[7:0]		18h	1Eh
58h	[0]		0	1
58h	[1]		0	1
5Ah	[0]		0	1
Chip ID = 8Bh				
E4h	[4:0]		15h	(Read Only)
E5h	[4:0]		15h	(Read Only)

*1 : Read Chip ID = 8Bh, Address = E4h, bit = [4:0] and set the value by subtracting "6" from that value.

Example) If Chip ID = 8Bh, Address = E4h, bit = [4:0] read value is "10h", the setting value is as follows.
Setting value = 10h - 6h = 0Ah

*2 : Read Chip ID = 8Bh, Address = E5h, bit = [4:0] and set the value by subtracting "6" from that value.

Example) If Chip ID = 8Bh, Address = E5h, bit = [4:0] read value is "20h", the setting value is as follows.
Setting value = 20h - 6h = 1Ah

5.3. Sensor setting flow and sequence

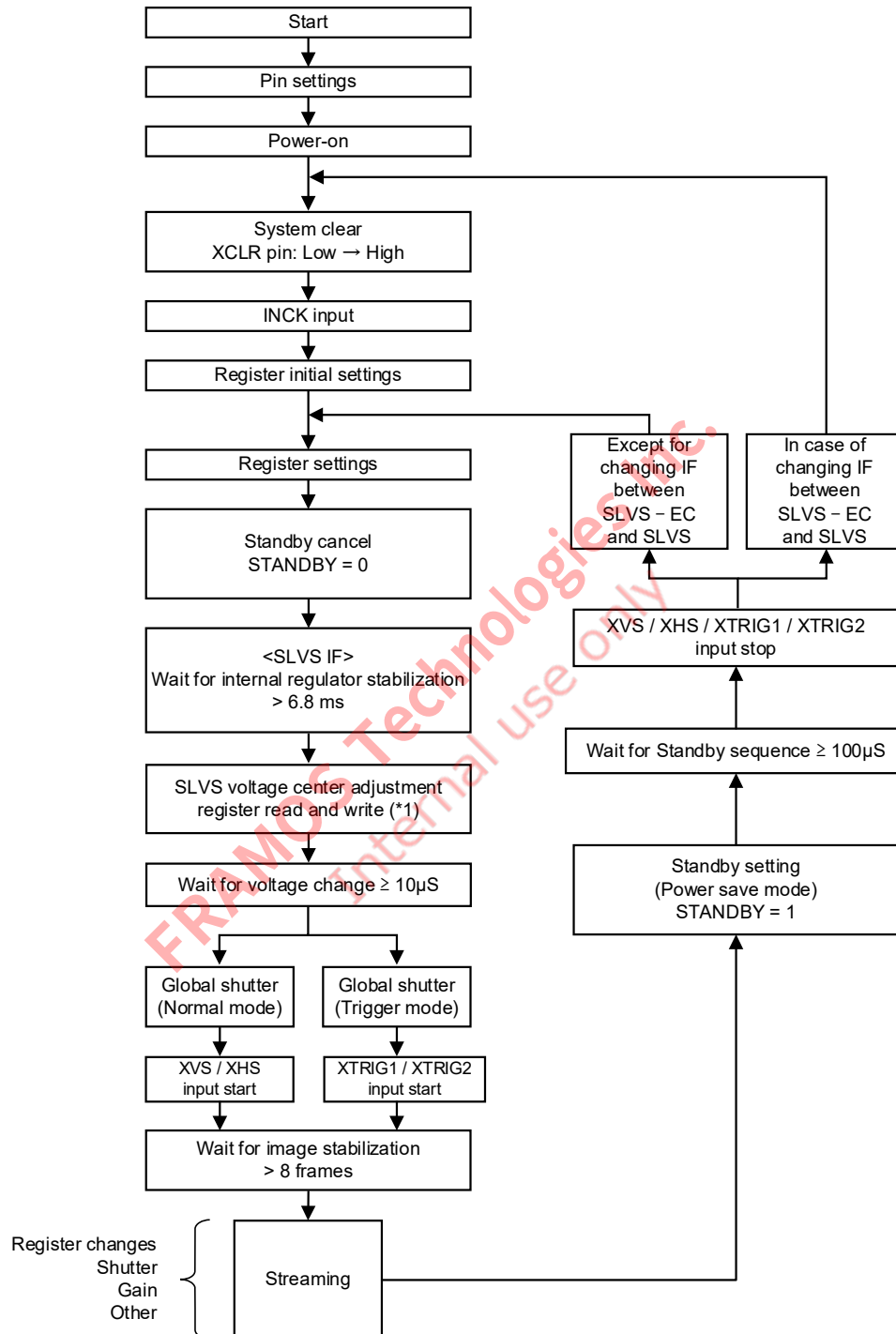
Chip ID = 8Bh, Address = E4h, bit = [4:0] and Chip ID = 8Bh, Address = E5h, bit = [4:0] will not change once read. Therefore, it is possible to adjust the SLVS voltage center using the following two methods.

- 5.3.1 Read the register value every time standby is canceled, and adjust the SLVS voltage center.
- 5.3.2 Adjust the SLVS voltage center with the register value read in advance.

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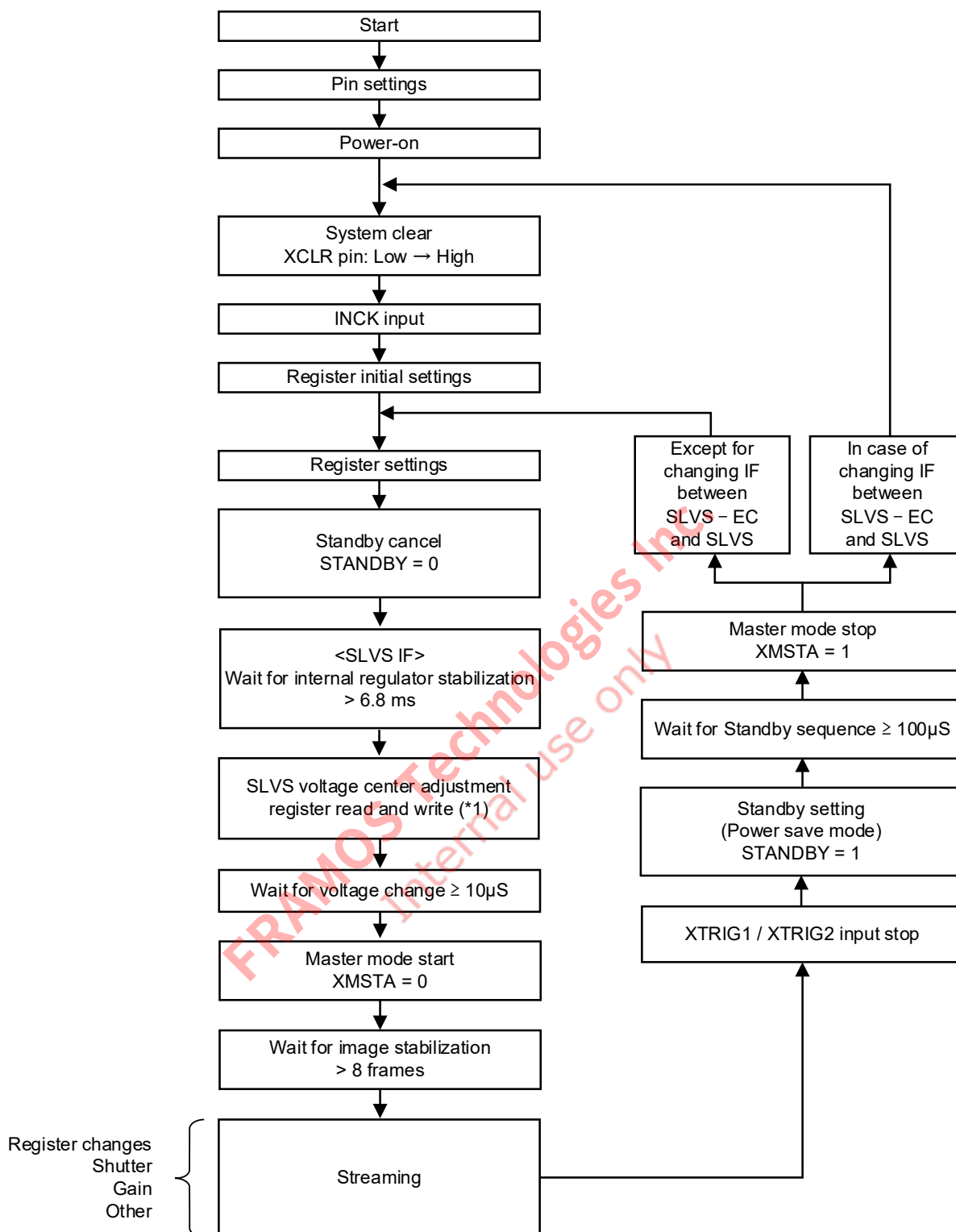
5.3.1. Read the register value every time standby is canceled, and adjust the SLVS voltage center.

The following describes how to read the register setting value each time standby is canceled and use the read value to adjust the SLVS voltage center. After "Wait for image stabilization", read and write the register that adjusts the SLVS voltage center.



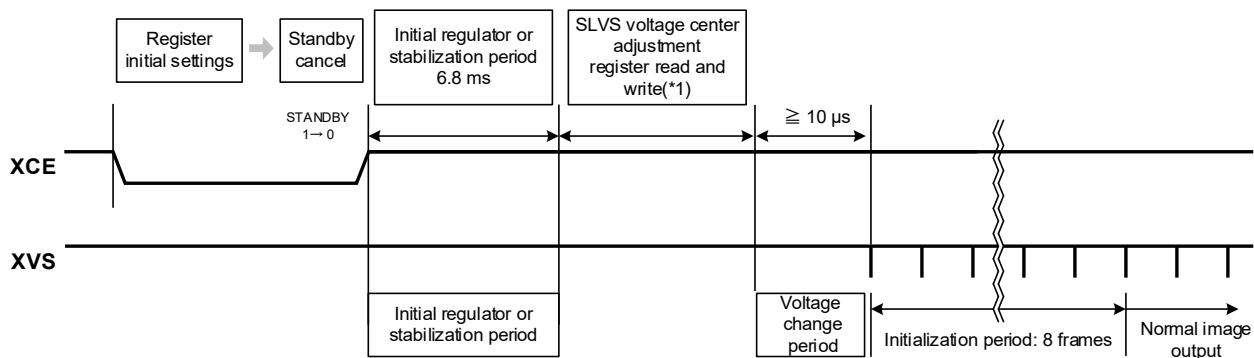
(*1) Register read: Chip ID = 8B, Address = E4h / E5h
 Register write: Chip ID = 0B, Address = 52h / 53h
 Chip ID = 0B, Address = 56h / 58h / 5Ah
 See the "Register Settings" section for details.

Sensor Setting Flow (Sensor Slave Mode)



(*1) Register read: Chip ID = 8B, Address = E4h / E5h
 Register write: Chip ID = 0B, Address = 52h / 53h
 Chip ID = 0B, Address = 56h / 58h / 5Ah
 See the "Register Settings" section for details.

Sensor Setting Flow (Sensor Master Mode)



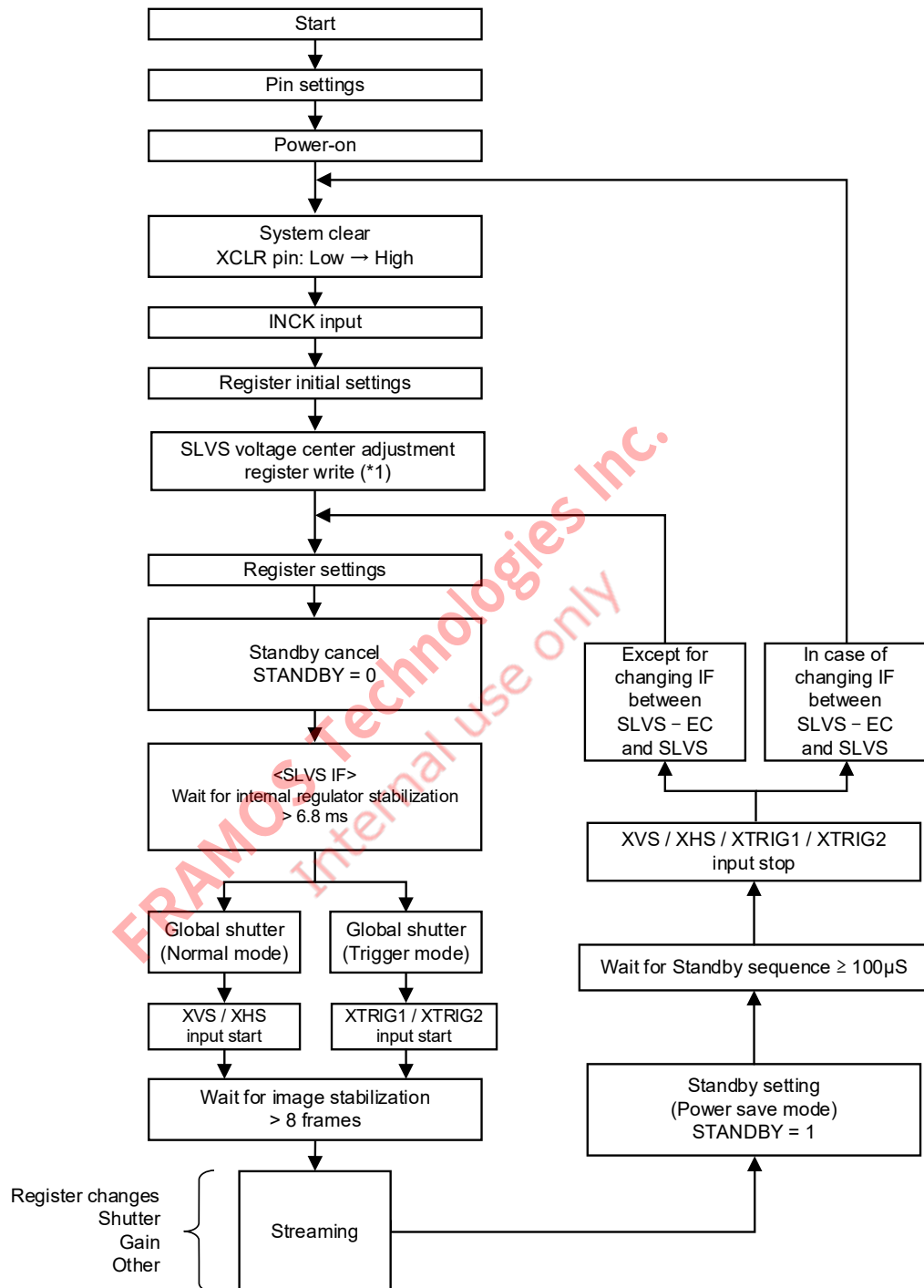
(*1) See the "Register Settings" section for details.

Sensor setting sequence

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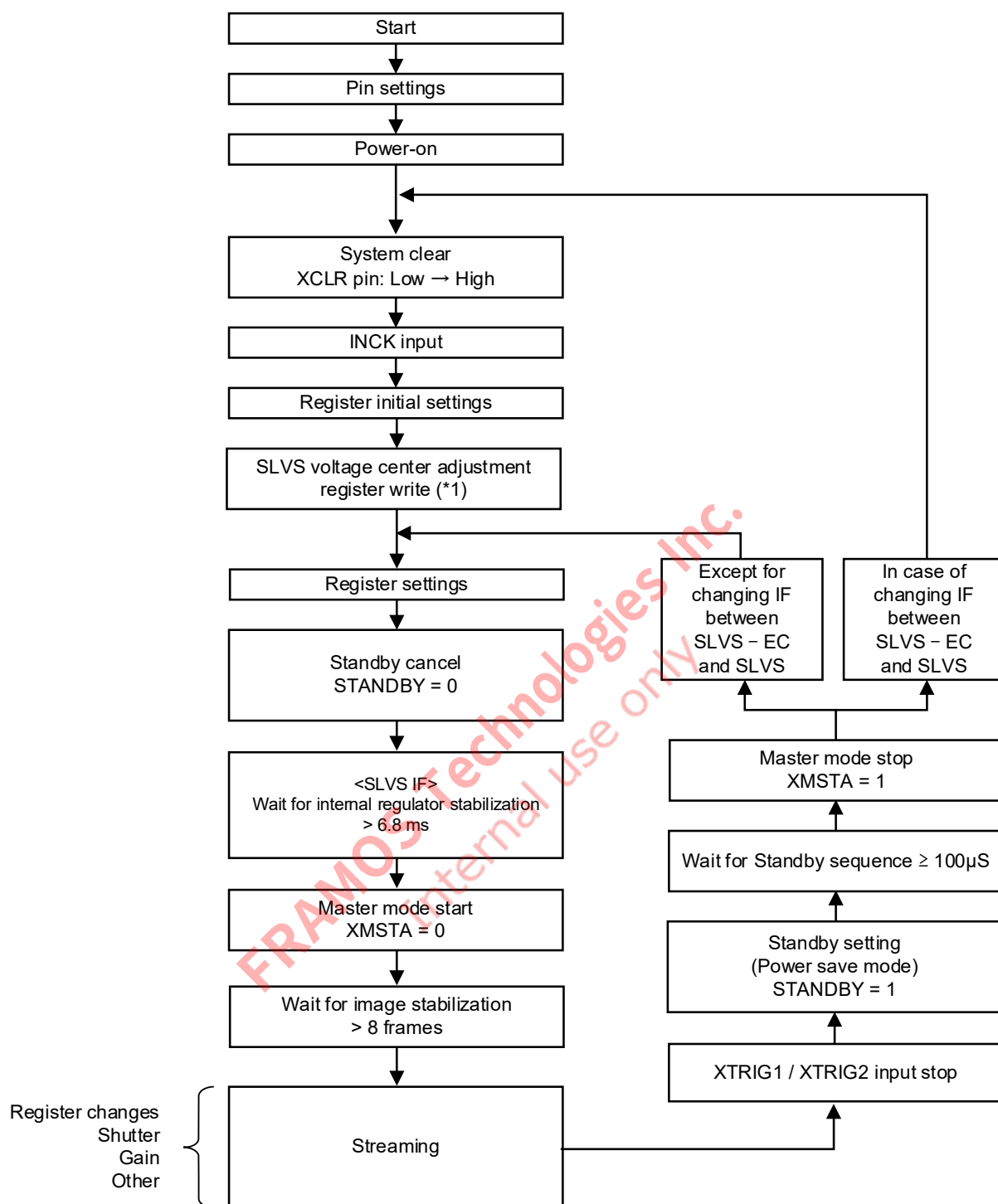
5.3.2. Adjust the SLVS voltage center with the register value read in advance.

Describes how to adjust the SLVS voltage center using the register setting value read in advance. After "Register initial settings", write the register that adjusts the SLVS voltage center.



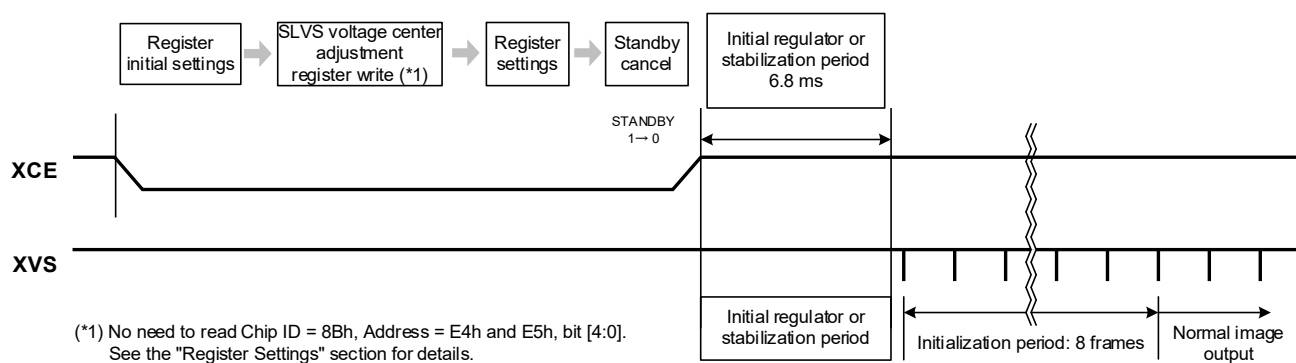
(*1) Register write: Chip ID = 0B, Address = 52h / 53h
Chip ID = 0B, Address = 56h / 58h / 5Ah
See the "Register Settings" section for details.

Sensor Setting Flow (Sensor Slave Mode)



(*1) Register write: Chip ID = 0B, Address = 52h / 53h
Chip ID = 0B, Address = 56h / 58h / 5Ah
See the "Register Settings" section for details.

Sensor Setting Flow (Sensor Master Mode)



Sensor setting sequence

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6. Note

This function is applicable only for SLVS IF. Not applicable for SLVS - EC IF and CSI-2 IF.

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Revision History

Version	Date	Page	Remarks
Rev.1.0	2023 / 2 / 14	–	First Edition
Rev.2.0	2023 / 3 / 10	6	Added that the maximum differential voltage rises by 100 mV and the VDDL pin current increases by about 10 mA maximum to the feature overview.
		6	Deleted Vcm value in "Definition of the characteristics of SLVS (Single end output)" figure
		6	Added Differential Voltage to "Definition of the characteristics of SLVS (Single end output)" table.

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